

Your grade: 100%Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.[Next item →](#)**1. What do you think applying this filter to a grayscale image will do?****1 / 1 point**

$$\begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

- ☐ Detecting image contrast.
- ☒ Detect 45-degree edges.
- ☐ Detect horizontal edges.
- ☐ Detect vertical edges.

 **Nice work**

Correct. Notice that there is a high delta between the values in the top left part and the ones in the bottom right part. When convolving this filter on a grayscale image, the edges forming a 45-degree angle with the horizontal will be detected.

2. Suppose your input is a 128 by 128 grayscale image, and you are not using a convolutional network. If the first hidden layer has 256 neurons, each one fully connected to the input, how many parameters does this hidden layer have (including the bias parameters)?**1 / 1 point**

- ☒ 4194560
- ☐ 12582912
- ☐ 12583168
- ☐ 4194304

 **Nice work**

Correct, the number of inputs for each unit is 128×128 since the input image is grayscale, so we need $128 \times 128 \times 256$ parameters for the weights and 256 parameters for the bias thus $128 \times 128 \times 256 + 256 = 4194560$.

3. Suppose your input is a 256 by 256 color (RGB) image, and you use a convolutional layer with 128 filters that are each 7×7 . How many parameters does this hidden layer have (including the bias parameters)?

1 / 1 point

- ☐ 6400
- ☐ 1233125504
- ☐ 18816
- ☒ 18944

 **Nice work**

Yes, you have $7 \times 7 \times 3 + 1$ weights per filter with the bias. Given that you have 128 filters, you get $(7 \times 7 \times 3 + 1) \times 128 = 18944$.

4. You have an input volume that is $121 \times 121 \times 16$, and convolve it with 32 filters of 4×4 , using a stride of 3 and no padding. What is the output volume?

1 / 1 point

- ☐ $118 \times 118 \times 16$
- ☐ $40 \times 40 \times 16$
- ☐ $118 \times 118 \times 32$
- ☒ $40 \times 40 \times 32$

 **Nice work**

Correct, using the formula $n_H^{[l]} = \frac{n_H^{[l-1]} + 2 \times p - f}{s} + 1$ with $n_H^{[l-1]} = 121$, $p = 0$, $f = 4$, and $s = 3$ we get 40

5. You have an input volume that is 61x61x32, and pad it using “pad=3”. What is the dimension of the resulting volume (after padding)?

1 / 1 point

- ☐ 64x64x35
- ☒ 67x67x32
- ☐ 61x61x35
- ☐ 64x64x32

5. You have an input volume that is 61x61x32, and pad it using “pad=3”. What is the dimension of the resulting volume (after padding)?

1 / 1 point

- ☐ 64x64x35
- ☒ 67x67x32
- ☐ 61x61x35
- ☐ 64x64x32

 **Nice work**

Yes, if the padding is 3 you add 6 to the height dimension and 6 to the width dimension.

6. You have an input volume that is 63x63x16, and convolve it with 32 filters that are each 7x7, and stride of 1. You want to use a “same” convolution. What is the padding?

1 / 1 point

- ☒ 3
- ☐ 2
- ☐ 1
- ☐ 7

 **Nice work**

Correct, you need to satisfy the following equation: $n_H - f + 2 \times p + 1 = n_H$ as you want to keep the dimensions between the input volume and the output volume.

7. You have an input volume that is 128x128x12, and apply max pooling with a stride of 4 and a filter size of 4. What is the output volume?

1 / 1 point

- ☒ $32 \times 32 \times 12$
- ☐ $32 \times 32 \times 3$
- ☐ $128 \times 128 \times 3$
- ☐ $64 \times 64 \times 12$

7. You have an input volume that is 128x128x12, and apply max pooling with a stride of 4 and a filter size of 4. What is the output volume?

1 / 1 point

- ☒ $32 \times 32 \times 12$
- ☐ $32 \times 32 \times 3$
- ☐ $128 \times 128 \times 3$
- ☐ $64 \times 64 \times 12$

 **Nice work**

Yes, using the formula $n_H^{[l]} = \frac{n_H^{[l-1]} + 2 \times p - f}{s} + 1$ with $p = 0$, $f = 4$, $s = 4$ and $n_H^{[l-1]} = 32$.

8. Which of the following are hyperparameters of the pooling layers? (Choose all that apply)

1 / 1 point

☒ Filter size. **Nice work**

Yes, although usually, we set $f = s$ this is one of the hyperparameters of a pooling layer.

☐ Average weights.☐ Number of filters.☒ Whether it is max or average. **Nice work**

Yes, these are the two types of pooling discussed in the lectures, and choosing which to use is considered a hyperparameter.

9. Which of the following are the benefits of using convolutional layers? (Check all that apply)

1 / 1 point

☒ It reduces the total number of parameters, thus reducing overfitting through parameter sharing.

☒ Whether it is max or average.

✔ **Nice work**

Yes, these are the two types of pooling discussed in the lectures, and choosing which to use is considered a hyperparameter.

9. Which of the following are the benefits of using convolutional layers? (Check all that apply)

1 / 1 point

☒ It reduces the total number of parameters, thus reducing overfitting through parameter sharing.

✔ **Nice work**

Yes, a convolutional layer uses parameters sharing and has usually a lot fewer parameters than a fully-connected layer.

☒ Convolutional layers are good at capturing translation invariance.

✔ **Nice work**

Yes, this is due in part to applying the same filter all over the image.

☐ It reduces the computations in backpropagation since we omit the convolutional layers in the process.

10. The sparsity of connections and weight sharing are mechanisms that allow us to use fewer parameters in a convolutional layer making it possible to train a network with smaller training sets. True/False?

1 / 1 point

☐ False

☒ True

✔ **Nice work**

Yes, weight sharing reduces significantly the number of parameters in a neural network, and sparsity of connections allows us to use a smaller number of inputs thus reducing even further the number of parameters.